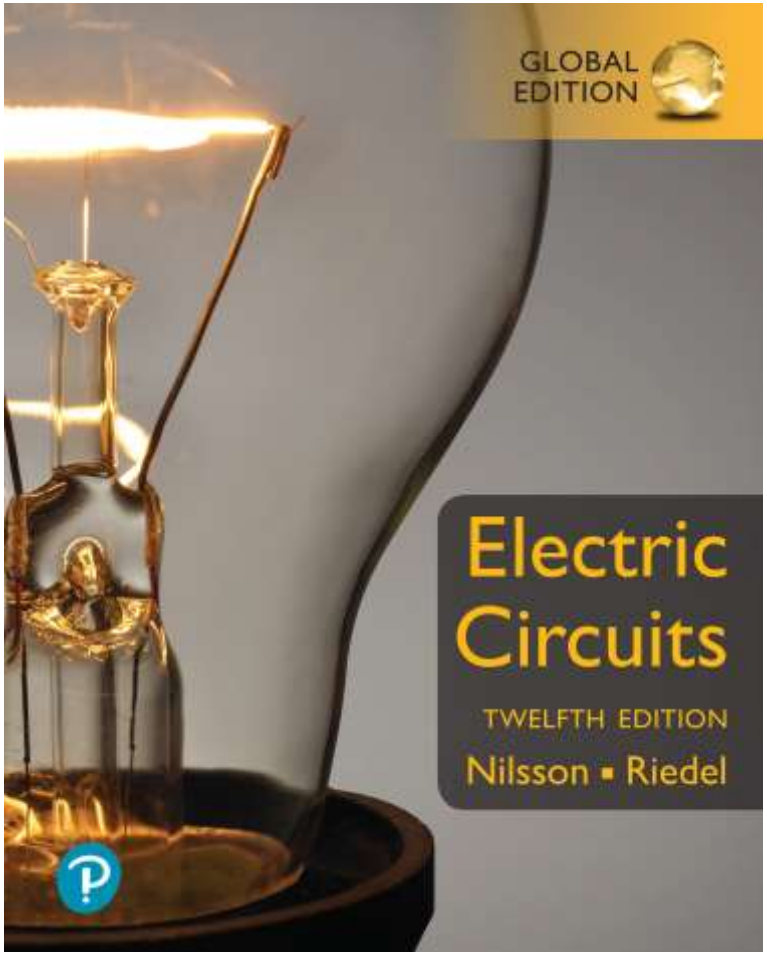
Electric Circuits

Twelfth Edition, Global Edition



Chapter 4 – Part 4

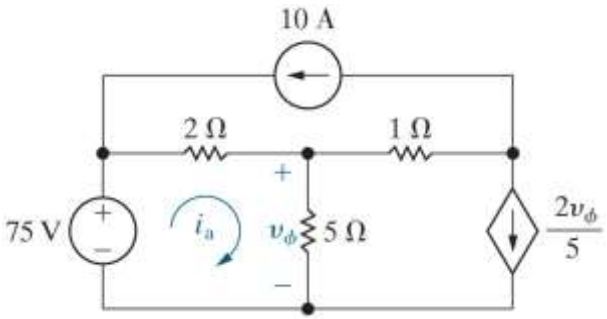
Techniques of Circuit Analysis

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (1 of 12)

What is the value of the mesh current in the upper mesh?



A. 10 A

B. −10 A

C. 75 V / 2 + 3

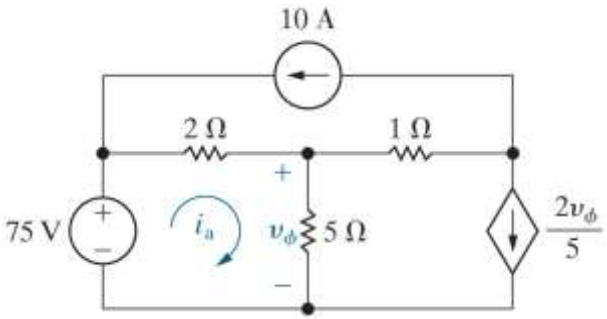
D. We cannot tell until we analyze the circuit!

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (1 of 12)

What is the value of the mesh current in the upper mesh?



- ✗ A. 10 A
- ✓ B. −10 A
- ✗ C. 75 V / 2 + 3
- ✗ D. We cannot tell until we analyze the circuit!

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (2 of 12)

Modify the mesh current analysis method to handle circuits with a current source in a single mesh:

- Step 1 – Identify the meshes with curved directed arrows that follow the perimeter of each mesh.
- Step 2 – Label the mesh currents for each mesh.
  - If a current source is in a single mesh, label the mesh current with the value of the current source.
- Step 3 – Write a KVL equation for each mesh where the current is unknown.
  - If there are dependent sources, write a dependent source constraint equation for each.
- Step 4 – Solve the KVL equations to find the mesh current values.
- Step 5 – Solve the circuit using mesh currents from Step 4 to find component currents, voltages, and power values.

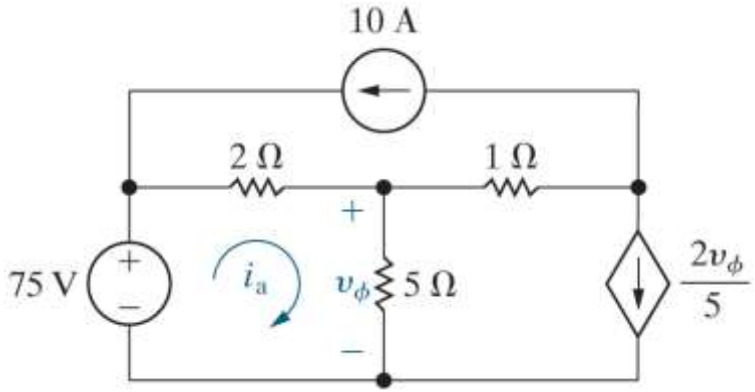
A current source in a single mesh is the dual of a voltage source between two nonreference essential nodes.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (3 of 12)

If we use the revised analysis method to label the mesh currents, how many KVL equations will we have to write?



A. 3

B. 2

C. 1

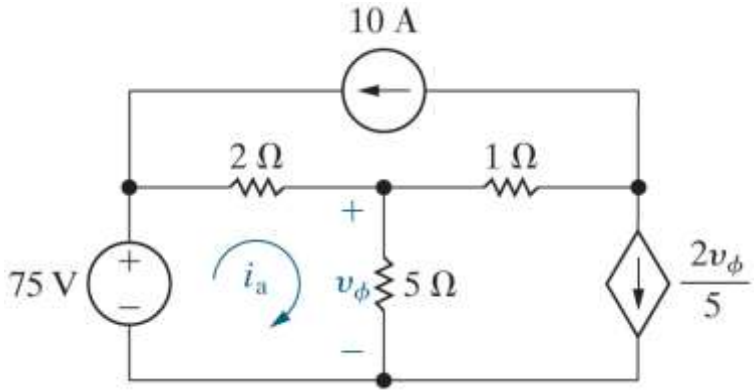
D. 0

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (3 of 12)

If we use the revised analysis method to label the mesh currents, how many KVL equations will we have to write?



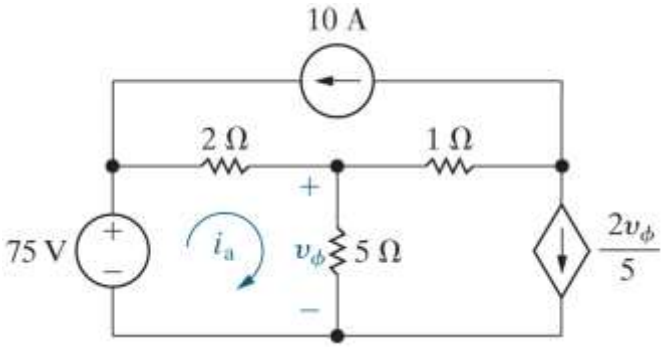
- ✗ A. 3
- ✗ B. 2
- ✓ C. 1
- ✗ D. 0

Pearson

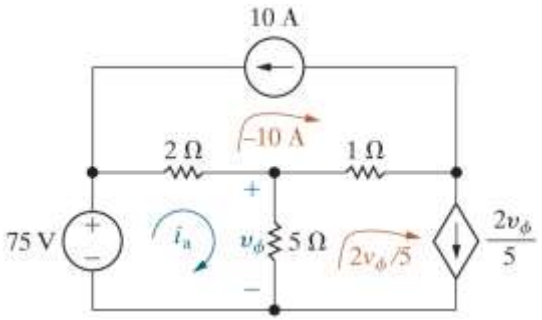
© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (4 of 12)

Find i_a using the mesh current method.



After Steps 1 and 2:



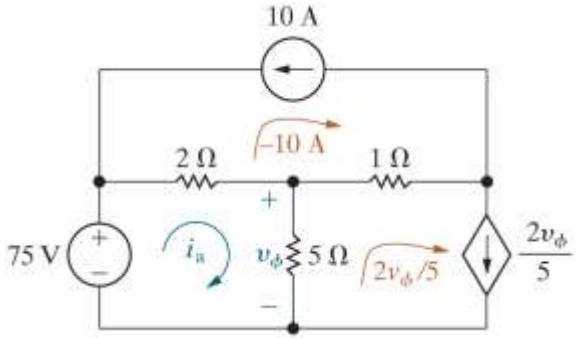
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (5 of 12)

Find i_a using the mesh current method.

Write the KVL equations and dependent source constraint equations (Step 3) and solve them (Step 4):



i_a mesh: $-75 + 2(i_a - (-10)) + 5\left(i_a - \frac{2v_\varphi}{5}\right) = 0$

constraint: $v_\varphi = 5\left(i_a - \frac{2v_\varphi}{5}\right)$

Std. form: $i_a(2+5) + v_\varphi(-2(5)/5) = 75 - 20$

$i_a(5) + v_\varphi(-2/5 - 1) = 0$

Solving: $i_a = 15$ A; $v_\varphi = 25$ V

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

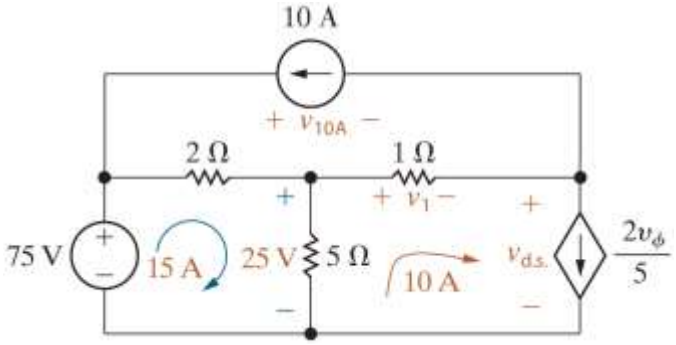
Chapter 4 – techniques of circuit analysis (6 of 12)

Power balance:

$$v_1 = (-10 - 10)/1 = -20 \text{ V}$$

$$v_{ds} = v_1 + v_\varphi = -20 + 25 = 5 \text{ V}$$

$$v_{10A} = 75 - v_{ds} = 75 - 5 = 70 \text{ V}$$



Component	Equation	p [W]
75 V	−(15)(75)	−1125
Dep. source	(5)(10)	50
10 A	−(70)(10)	−700
2Ω	$(15+10)^2(2)$	1250
1Ω	$(10+10)^2(1)$	400
5Ω	$25^2/5$	125

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (7 of 12)

The node voltage method has a “special case” when a circuit has a supernode – a voltage source that is the only component in a branch connecting two nonreference essential nodes. The mesh current method has a special case for “super meshes”, which exist when

A. A voltage source is in a single mesh.
B. A current source is in a single mesh.
C. A voltage source is shared between two meshes.
D. A current source is shared between two meshes.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (7 of 12)

The node voltage method has a “special case” when a circuit has a supernode – a voltage source that is the only component in a branch connecting two nonreference essential nodes. The mesh current method has a special case for “super meshes”, which exist when

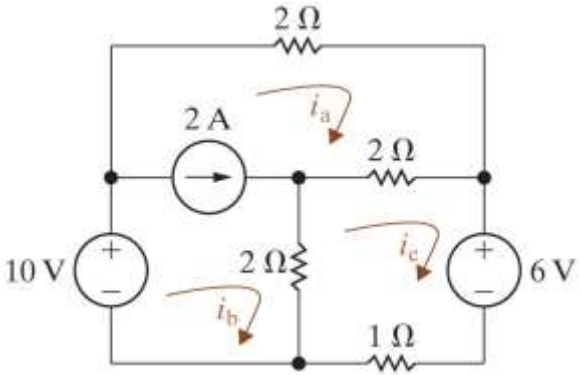
- ✗ A. A voltage source is in a single mesh.
- ✗ B. A current source is in a single mesh.
- ✗ C. A voltage source is shared between two meshes.
- ✓ D. A current source is shared between two meshes.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (8 of 12)

This circuit has a supermesh created from the two meshes with the following mesh currents:



A. i_a and i_b

B. i_a and i_c

C. i_b and i_c

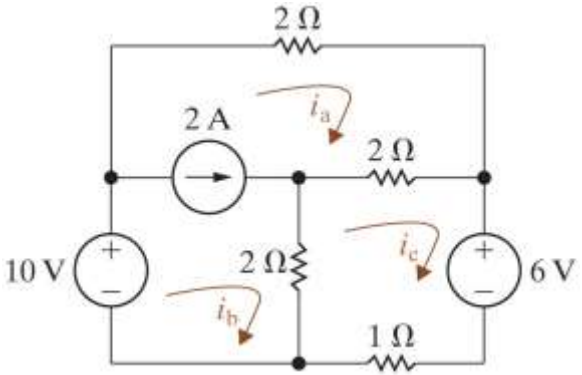
D. i_a and i_b and i_c

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (8 of 12)

This circuit has a supermesh created from the two meshes with the following mesh currents:



- ✔ A. i_a and i_b
- ✘ B. i_a and i_c
- ✘ C. i_b and i_c
- ✘ D. i_a and i_b and i_c

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (9 of 12)

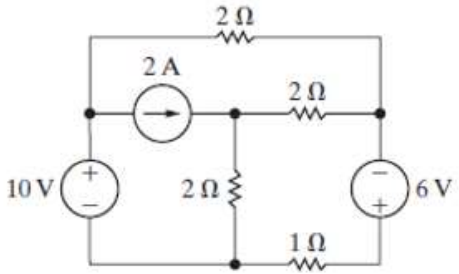
The final version of the mesh current analysis method, which handles all three special cases:

- Step 1 – Identify the meshes with curved directed arrows that follow the perimeter of each mesh.
- Step 2 – Label the mesh currents for each mesh.
- If a current source is in a single mesh, label the mesh current with the value of the current source.
  - If a current source is shared between two meshes, combine the meshes to create a supermesh and mentally erase the current source.
- Step 3 – Write a KVL equation for each supermesh and for each single mesh where the current is unknown.
  - If there are dependent sources, write a dependent source constraint equation for each.
  - If there are supermeshes, write a supermesh constraint equation for each.
- Step 4 – Solve the KVL equations to find the mesh current values.
- Step 5 – Solve the circuit using mesh currents from Step 4 to find component currents, voltages, and power values.

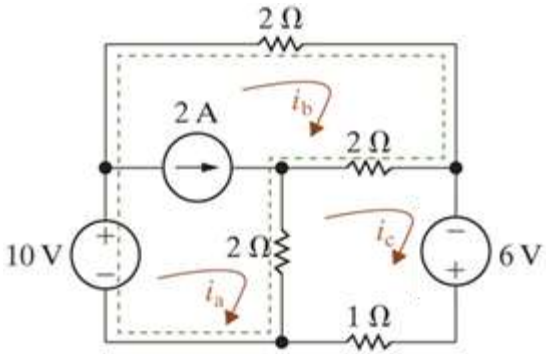
© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (10 of 12)

Find the power absorbed by the $1\,\Omega$ resistor using the mesh current method.



After Steps 1 and 2:

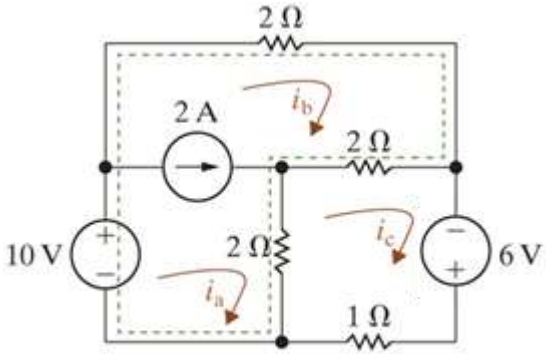


Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (11 of 12)

Writing the equations (Step 3) and solving them (Step 4).



supermesh: $-10 + 2i_b + 2(i_b - i_c) + 2(i_a - i_c) = 0$

i_c mesh: $-6 + 1i_c + 2(i_c - i_a) + 2(i_c - i_b) = 0$

SM constraint: $2 = i_a - i_b$

$$i_a(2) + i_b(2+2) + i_c(-2-2) = 10$$

Std form: $i_a(-2) + i_b(-2) + i_c(1+2+2) = 6$

$$i_a(1) + i_b(-1) + i_c(0) = 2$$

Solving: $i_a = 7$ A; $i_b = 5$ A; $i_c = 6$ A;

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Copyright



This work is protected by United Kingdom copyright laws and is provided solely for the use of instructors in teaching their courses and assessing student learning. Dissemination or sale of any part of this work (including on the World Wide Web) will destroy the integrity of the work and is not permitted. The work and materials from it should never be made available to students except by instructors using the accompanying text in their classes. All recipients of this work are expected to abide by these restrictions and to honor the intended pedagogical purposes and the needs of other instructors who rely on these materials.

© 2024 Pearson Education Ltd. All Rights Reserved.